High Contrast Imaging

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Keywords

Optics, coronagraphs, exoplanets, computational methods

Abstract

High Contrast Imaging will enable the direct detection of photons from rocky terrestrial worlds.

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HCI: High Contrast Imaging

FP: Focal Plane

PP: Pupil Plane

FPWFS: Focal Plane Wavefront Sensing

WFS: Wavefront Sensor

1. INTRODUCTION

In this review, we provide interactive notebooks that enable the reader to build up their intuition on how coronagraphs work along with wavefront sensing both in the pupil and the focal planes, and how we will tackle the challenges in reaching the contrasts of 10^{-10} at angular separations of less than one arcsecond that are required.

This manuscript was prepared using the showyourwork package[1] and the source code used to generate each figure is available in a public GitHub repository[2].

To see the specific version of the Jupyter notebook, that was executed to generate each figure, click on the icon next to the figure caption.

2. Science Goals

Initially developed to image the Sun's corona without the need for a Solar eclipse, one of the most significant science drivers for the latest coronagraphs is in the detection and characterisation of circumstellar material and planets around nearby stars. Planetary systems that are closer to the Sun have two benefits: given two identical planetary systems, one that is twice as close as the other will have twice the angular separation between the star and planet, and from the inverse square law, four times more flux is received from the

[1] https://show-your.work
[2] https://github.com/mkenworthy/ARAA_HCI

planet.

For direct imaging, therefore, the closest stars to the Sun are the ones that are studied for the presence of directly imaged exoplanets. In a volume limited (20 parsecs) sample of stars around the Sun (Kirkpatrick et al. 2023), there are XXXX solar (G2) type stars, and YYYY M-dwarf stars. DI young Jupiters look are self-luminous in the NIR, but for terrestrial planets, looking for light reflected from the atmosphere/surface, so subject to inverse square law for planet separation from star.

Two regimes: solar analogues at optical wavelengths lead to space based missions, ground based larger aperture telescopes study M dwarf stars that are on average closer to the Sun, and look in the NIR.

Mention HABCAT for a catalogue of solar like stars.

Bioactive molecules see reviews of Seager et al. (2016); Kaltenegger (2017); Schwieterman et al. (2018).

3. Brief history

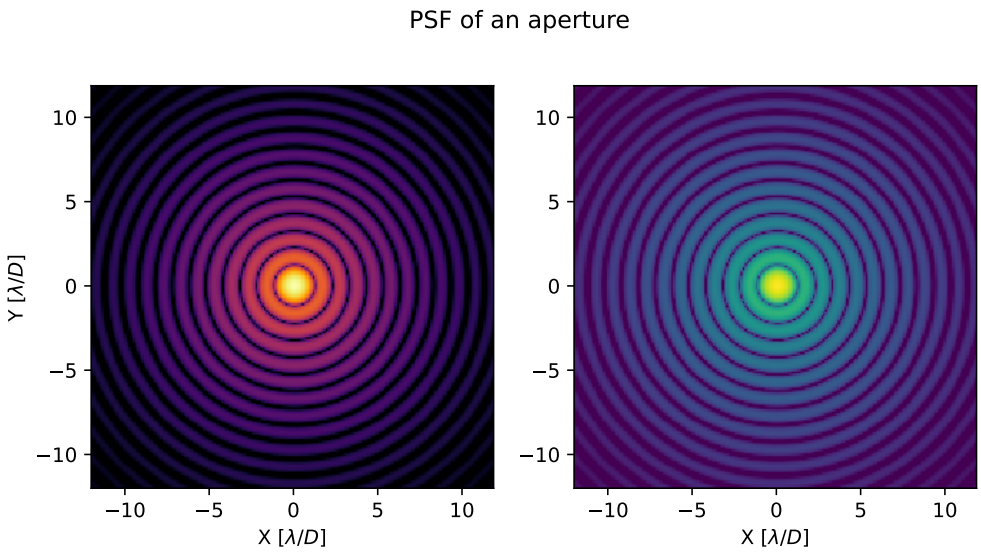

Figure 1

Figure run from a static Python script.

The scientific goal is to image an exoplanet (a planet orbiting a star other than our Sun) and to characterise its orbit and atmosphere. For an Earth analogue with similar radius, albedo and effective temperature orbiting around a solar-type star 10 parsecs away, the typical amount of reflected light in the optical is 10^{-10} of the central star at a separation of 0.1 arcseconds at maximum elongation. The technical challenge is in distinguishing the light of the parent star from the light of the planet.

3.0.1. From Maxwell's Equations to Wavefronts. The vast majority of energy from astrophysical objects arrives at our telescopes in the form of electromagnetic radiation. The electromagnetic waves propagate through space by exchanges between the electric and magnetic fields because a changing electric field induces a magnetic field and vice-versa. This time-dependent interaction is described by Maxwell's equations,

$$\begin{aligned}
\frac{\partial \mathcal{D}}{\partial t} &= \nabla \times \mathcal{H}, && \text{(Faraday's law)} \\
\frac{\partial \mathcal{B}}{\partial t} &= -\nabla \times \mathcal{E}, && \text{(Ampere's Law)} \\
\nabla \cdot \mathcal{B} &= 0, && \text{(Gauss's law)} \\
\nabla \cdot \mathcal{D} &= 0. && \text{(Coulomb's law)}
\end{aligned} \tag{1.}$$

This form of Maxwell's equations is in the material form without any charge and current sources. The material form is used to describe the propagation of electromagnetic fields inside matter. This set of equations is completed by describing the particular matter of the medium with the constitutive relations, $D = \epsilon E$ and $B = \mu H$. Here ϵ and μ are the permittivity and magnetic permeability, respectively. The wave equation for electromagnetic waves can be derived by taking the curl of Ampere's law. This gives us the classic wave equation if we assume that the EM wave propagates in isotropic and homogeneous materials,

$$\mu\epsilon \frac{\partial^2 \mathcal{E}}{\partial t^2} - \nabla^2 \mathcal{E} = 0. \tag{2.}$$

Let's now consider a purely monochromatic EM wave with frequency ω. The wave function is then $\mathcal{E} = \psi(r)e^{iwt}$ with $\psi(r)$ describing the spatial distribution. Substituting this relation into Equation 2,

$$\nabla^2 \mathcal{E} + \mu\epsilon\omega^2 \mathcal{E} = 0. \tag{3.}$$

The usual definition of the permittivity and permeability are $\epsilon = \epsilon_r \epsilon_0$ and $\mu = \mu_r \mu_0$ with ϵ_r and μ_r the relative permittivity and permeability compared to that of vacuum, ϵ_0 and μ_0. In optics most glasses and materials are defined by their refractive index n which is related to permittivity as $\epsilon_r = n^2$ and many of these are also non-magnetic which means that $\mu_r = 1$. Substituting this will give us the classic Helmholtz equation

$$\nabla^2 \mathcal{E} + n^2 k^2 \mathcal{E} = 0, \tag{4.}$$

We made use of the fact that the speed of light is $c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$ and that $k = c\omega$ is the wave number. The propagation through an optical system has a preferential direction that is usually defined along the z-axis. The z-axis evolution can be derived by separating the spatial components into the z component and the perpendicular components (x,y),

$$\frac{\partial^2 \mathcal{E}}{\partial z^2} = -\nabla_\perp^2 \mathcal{E} - n^2 k^2 \mathcal{E}, \tag{5.}$$

This differential equation can be solved by assuming a plane wave expansion $\mathcal{E}(x, y, z) = e^{i(k_x x + k_y y)} f(z)$ which results in

$$\frac{\partial^2 \mathcal{E}}{\partial z^2} = -(n^2 k^2 - k_\perp^2)\mathcal{E}. \tag{6.}$$

The solution to this equation is the so called Angular Spectrum Propagator that relates the electric field at any one plane to the electric field at any other,

$$\mathcal{E}(x, y, z') = \mathcal{F}_{x,y}^{-1}\{e^{-ik_z(z'-z)} \mathcal{F}_{x,y}\{\mathcal{E}(x, y, z)\}\}. \tag{7.}$$

Here the z component of the wave vector is defined as $k_z = \sqrt{n^2 k^2 - k_\perp^2}$ and $\mathcal{F}_{x,y}^{(-1)}$ is defined as the (inverse) Fourier transform over the x and y coordinates. While Equation 7 describes the full propagation from one plane to another, it is quite unwieldy to use and does not provide much physical insight. For many

optical systems it is sufficient to analyze the paraxial performance. The paraxial approximation assumes that the plane waves make small angles with respect to the z-axis which means that the x and y wave vector components are $k_x, k_y << 1$. In this regime the propagation factor $k_z = \sqrt{n^2k^2 - k_\perp^2} \approx nk - \frac{k_\perp^2}{2nk}$

$$\mathcal{E}(x, y, z') = e^{-ink(z'-z)}\mathcal{F}_{x,y}^{-1}\{\mathcal{F}_{x,y}\{\mathcal{E}(x, y, z)\}\}. \quad 8.$$

The coherent superposition of electric fields from different parts of the telescope aperture and the associated phase offset gives rise to **interference**, where the sum of all these electric amplitudes can give very different results from an incoherent addition of the fluxes.

4. Definition of quantities

5. Diffraction of simple apertures

[Notebook: different apertures and geometries]

A square aperture with a side length of D illuminated by light of wavelength λ will produce the illumination seen in Figure XXXX. The focal plane is divided by a rectilinear dark grid The angular distance from the optical axis to the first dark minimum is λ/D in units of radians.

For a circular unobstructed telescope aperture, this pattern is referred to as the Airy function.

6. The Lyot Coronagraph

[Notebook: lyot fully adjustable] Stellar coronagraphs have now been in use for several decades. However, the first coronagraph was developed by Bernard Lyot to observe the corona of the Sun in the 1930s. It took over half a century before astronomers applied coronagraphs to image the faint cirumstellar environment by blocking starlight.

The first coronagraph to successfully image a debris disk was a Lyot coronagraph built by Vilas & Smith (1987) and it imaged the edge-on circumstellar disk around Beta Pictoris in 1984 (Smith & Terrile 1984). The optical layout of the Lyot coronagraph can be generalised in Figure XXXXX, with the letters A-F representing the images present at that location in the coronagraph light path. The telescope pupil (A) is reimaged into a focal plane of the sky (B) where an opaque mask that has high absorptivity and low reflectivity blocks the light from any on-axis source (C). Optics then form an image of the resultant pupil (D) to an intermediate pupil plane, where a second opaque mask - the Lyot stop - is located. This Lyot stop blocks the diffracted light of the star - now forming a ring of light along the outer edge of the telescope pupil (E). A second optical system then reimages the resultant focal plane of the instrument onto a detector (F). Any circumstellar objects outside the radius of the focal plane mask then pass through unimpeded through the coronagraph and are subsequently reimaged in the detector focal plane at F. The light rays pass through the coronagraph optics to form an image at F with only minor modification: the reimaged pupil at D is superficially very similar to the telescope pupil A.

For an on-axis source, the removal of the Airy core plus attendant diffraction rings significantly modifies the wavefront passing through the coronagraph, resulting in a flux redistribution at D where the flux is concentrated in a ring whose peak brightness lies along the perimeter of the reimaged telescope pupil, extending both beyond the radius of the pupil and into the centre of the pupil.

The purpose of the Lyot stop is to remove as much of this ring of light as possible, whilst maximising the throughput of the pupil image D for off-axis sources. Adjusting the diameter of the focal plane mask changes the full width half maximum of the ring of light at D, which requires a smaller Lyot stop to block - but then the throughput of the pupil for off-axis sources then decreases. Decreasing the Lyot stop aperture has a second impact in that the reduced pupil diameter increases the Full Width Half Maximum (FWHM)

of the images in the final focal plane F, spreading the flux from the off-axis sources over a larger area in the detector and degrading the angular resolution of the telescope and instrument. The optimal diameters of the Focal Plane Mask (FPM) and Lyot stop aperture are then driven by the science requirements - how close to the central star (measured in diffraction widths at the lower spatial resolution) should the coronagraph be able to transmit light from off-axis objects in the field of view.

7. Focal plane phase mask coronagraphs

The Lyot coronagraph was designed at the time when it was still difficult to precisely manipulated the phase of light with optics. With the advent of more advanced manufacturing capabilities, very precise phase control became possible. This was a significant boost to the design toolbox for coronagraphs. The major downside of the classic Lyot coronagraph is that only the light that falls on the opaque focal plane masks gets blocked. Any light that is off-axis will pass through the system. This holds not only for light that comes from off-axis sources but also for aberrations that causes light to end up outside of the mask. A bigger mask will be able to block a larger fraction of the light and therefore achieve a deeper contrast. However, if the mask is made larger the inner-working angle also becomes bigger and that means fewer planets will be accessible. A smaller inner-working angle is crucial both for the extremely large ground-based telescopes and the space-based telescopes.

Focal plane phase masks offer a solution. Phase masks don't enhance the contrast by blocking light but by phase shifting a part of the PSF (usually the core of the PSF). The phase shifted part then causes destructive interference at the Lyot plane. The Lyot stop blocks the areas where the light does not destructively interferes. The 50% encircled energy radius is on the order of $\sim \lambda/D$ for almost all aperture shapes. This means that it is possible to achieve perfect destructive interference with a mask that has a size on the order of λ/D. This is the central idea that was used to design the Roddier and Roddier (RR) phase mask Roddier & Roddier (1997). The RR mask covers the core of the Airy pattern that contains 50% of the encircled energy and phase shifts the core by π to cause destructive interference. The RR mask works well for monochromatic light. However, over broad spectral bandwidths it degrades in contrast. Diffraction causes wavelength scaling of the PSF that either makes the focal plane mask too big or too small compared to the PSF. Masks made out of multiple concentric rings, such as the dual-zone phase mask Soummer et al. (2003), were proposed to increase the spectral bandwidth.

Further achromatization for larger spectral bandwidths is possible by designing inherent achromatic phase masks. Inherent achromatic masks are scale invariant, which means that they always look the same regardless of the size of the PSF. One of the most used phase mask coronagraphs is the vortex coronagraph (VC). The VC uses a phase mask with a vortex pattern where the phase changes with the azimutal angle: $\phi = q \cdot \theta$ with q the charge, i.e. the number of times the phase wraps around, of the vortex and θ the azimuth angle. Another example is the Four Quadrant Phase Mask (FQPM). The FQPM splits the focal plane in four quadrants and applies a checkerboard phase pattern. The scale invariant phase masks of these two coronagraphs are shown in Figure 2.

Both phase masks that are shown in Figure 2 completely null out all on-axis light from a clear aperture. This property is why both the FQPM and the OVC have been extensively studied over the past several decades. The inner-working angle of both coronagraph approaches 1 λ/D which makes coronagraphy possible at the diffraction limit! The FQPM has been implemented on SPHERE [CITE SPHERE PAPER] and is currently the coronagraph on JWST with the smallest inner-working angle and has been recently used to characterize a planet at 1.8 λ/D Franson et al. (2024).

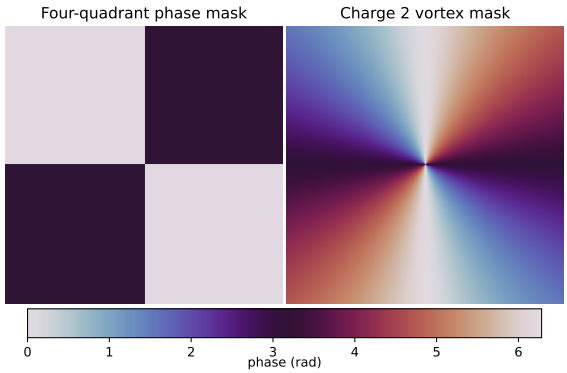

Figure 2

The focal plane phase functions of two scale invariant coronagraphs. The left figure shows the four quadrant phase mask and the right figure shows the optical vortex phase mask.

8. Parameters to optimize

As the previous section explained, clear apertures have solutions that perfectly remove any on-axis light. However, real systems are not ideal and generally don't have a clear aperture. Even more important, stars are not ideal point sources. Solutions that only work for point sources are already out of the question then. The current and future generation of coronagraphs are designed to take on non-ideal environments. This means that the coronagraph is optimized for a specific list of parameters during the design process.

The first set of parameters are the inner-working angle (IWA) and the outer-working angle (OWA). The IWA is usually set to the angular separation where the throughput is 50% of the peak off-axis throughput. The IWA and OWA set the smallest and largest angular separation of the dark hole region that the coronagraph will make. Some coronagraphs, such as the OVC or FQPM, don't have an OWA only an IWA. Figure 3 shows a coronagraphic dark hole with the definition of the IWA and OWA.

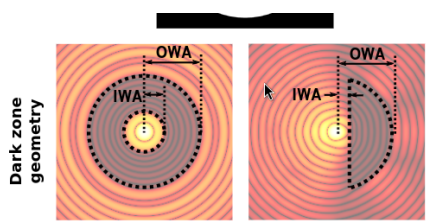

Figure 2. The definition of all masks used in this work. These masks are used for the parameter study in Sections 3 and 4. Centered masks are used for both point-symmetric and one-sided dark zones. The left-justified masks are for two-sided dark zones, while the right-justified masks are used for one-sided dark zones. In general though, arbitrary telescope pupils, Lyot masks, focal-plane masks and dark zone geometries can be used with a PAPLC.

Figure 3

ADD caption TODO: replace with our figure instead of Emiel's

The contrast and throughput are the next set of parameters that we need to optimize for. The contrast sets the amount of starlight that is left after the coronagraph. It is important to define the term contrast

as this can mean many different things. Ruane et al. provide a thorough overview of the different metrics and their definition. The contrast is defined as

$$C = \frac{\eta_*(\vec{r})}{\eta_p(\vec{r})}. \tag{9.}$$

Here $\eta_*(\vec{r})$ is the fractional throughput of the star at focal plane position $\vec{r}$ integrated over a photometric aperture. This is then divided by $\eta_p(\vec{r})$ the fractional throughput of the planet in the same photometric aperture. This normalizes the contrast w.r.t. the throughput of the planet, which is important because the planet throughput usually varies as function of angular separation. Both the contrast C and $\eta_p(\vec{r})$ need to be included in the optimization process. The first to make sure that the starlight is nulled and the second to make sure that the planet light is maintained. This optimization has to be done over a certain spectral bandwidth $\Delta\lambda$.

The corongraphs that are designed with only the previous set of optimization targets are not optimal in real environments. In real instrument environments there are wavefront aberrations and small instrumental drifts. These cause light to leak around the coronagraph and cause residual stellar speckles. The coronagraphs must be made robust against low-order wavefront errors and other instrumental drifts. Other more practical things to consider are the precision with which we can align an instrument. For example, how well can the Lyot Stop be aligned? The performance of a coronagraph might be extremely sensitive to the Lyot stop alignment, which means that theoretically the coronagraph delivers the contrast but practically it will never reach it. Therefore, alignment tolerancing must be included in the coronagraph design to make sure the target performance is achieved.

In this way, there are many other nuisance parameters that can be included. However, the numerical optimization will take significantly longer if more parameters are included. A good coronagraph designer will therefore make a trade-off between which parameters are required, good to have and not significant.

9. Direct Imaging of Terrestrial Exoplanets

TODO MAK

The conceptual design of GMagAO-X by Males et al. (2022).

Lives of speckles by Males et al. (2021).

Jared's AO4 ELT paper was a good talk, but this is the argument to limit to 30 parsec.

Fundamental sensitivity limite on apertures with PIAA-ZWFS on Haffert et al. (2023b)

Haffert 2023 talk mentioned this ore we can talk about it below:

Diffraction limit of telescope is almost fundamental limit. The HZ stays fixed in linear space, but the angular size shrinks with distance of star from Sun.

Apparent magnitude of the stars goes down, AO performance goes down, halo fills in.

Extreme Adaptive Optics ARAA in Guyon (2018)

REFS seager, keltenegger, 2018 paper from seager student about false negatives for detecting life, we need multiple signatures. ??? is it this paper? "Molecular simulations for the spectroscopic detection of atmospheric gases" by Sousa-Silva et al. (2019).

PIAA principle by Guyon (2003)

Guoyn 2006, 2010, 2014

Guoyn 2005 fundamental WFS in Guyon (2005)

Jared and Guyon 2018/19 on predictive control and speckle lifetime - (Males & Guyon 2018)

Jared 2020 speckle lifetime (Males et al. 2021)

self coherent camera (Pierre? Baudoz 2006 original (Baudoz et al. 2006)) refael Galicher paper from 2010 instead/as well Self-coherent camera as a focal plane wavefront sensor: simul (Galicher et al. 2010).

2022 on sky EFC axel potier (Potier et al. 2022) and the first paper in the series on VLT/SPHERE is (Potier et al. 2020) and on sky efc with FQPhase Mask to demo on-sky EFC is paper III and is Galicher et al. (2024).

The search for life beyond the Earth has focused on the idea that water is an essential part of life cycles elsewhere in the Universe, as it is a polar solvent formed from two elements that are found in abundance throughout the Galaxy. Places where water can exist in its liquid state form prime locations for these searches, notably Earth-like planets and water moons that have liquid water underneath an ice layer. The region around a star where liquid water can exist on the surface of a terrestrial planet (with appropriate atmospheric pressure) is referred to as the Habitable Zone (HZ); for the Sun this is from 0.9-1.2 au. THe HZ changes with stellar evolution as the luminosity of stars changes with time. The angular size of the HZ in diffraction widths can be approximated by XXXXX.

This combined with the greater frequency of lower mass stars means that there are two regimes for imaging HZ terrestrial planets around nearby stars: M dwarfs and G-type stars.

For solar type stars, the contrast required in the optical wavelengths is on the order of 10e-10 for terrestrial planets in the HZ of solar type stars, but for smaller mass stars with lower luminosities, the contrast is 1e-7 for M dwarfs since the similar sized planets subtend larger amounts of radiated flux.

10. ELTs for M dwarf exoplanets

TODO MAK

Guyon has a nice figure... 2014 or 2016 SPIE proceeding. Maybe Emiel's phd thesis introduction on this.

Cite papers about METIS (Brandl et al. 2021) has high contrast imaging modes (Kenworthy et al. 2016; Carlomagno et al. 2020).

PCS roadmap paper (Kasper et al. 2021)

GMagAOX from Jared Males 2022/2024 (Males et al. 2022).

TMT papers on the PSI Jensen-Clem et al. (2022); Fitzgerald et al. (2022)

XXX put an approximate outer limit on the distance for direct imaging. HZ of solar star is

$DL_{HZ} = 1/$

[Notebook: play with different contrasts and IWA to see yields]

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11. Space coronagraphs and HWO for G type stars

Chris Stark Space telescope yield papers...(Stark et al. 2014, 2024).

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Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

12. Polarization effects

[Notebook: Polarimetric beam shift using Fresnel equation - Change angle of incidence and show differential beam shift.]

The field of high-contrast imaging is always hammering away at one noise floor after another. It started with the common phase aberrations that dominate at low to moderate contrast. After that, amplitude aberrations started to limit the contrast. This was solved by using multiple deformable mirrors. Now, the contrasts that are achieved on-sky and on testbeds reveal another limit; polarization. Polarization is an often underappreciated property of light. The derivation shown in Section **??** actually also ignores the effects of polarization which was done for mathematical clarity. However, light consists of two orthogonal polarizations states that do not interfere with each other. That means that there are, at any time, always two beams of light propagating through our coronagraph that might interact in a different way with the instrument. A more detailed treatment of polarization and physical optics propagation can be found in the literature (McLeod & Wagner 2014).

The Fresnel equations describe how light is either reflected off or transmitted through an interface. The equations also show that different polarization states have different Fresnel coefficients. This wouldn't be a problem if the Fresnel equations did not also depend on the angle of incidence. The definition of all variables for the incidence, reflected and transmitted wave are shown in Figure 4.

The Frensel equations for plane wave interfaces are,

$$r_s = \frac{n_1 \cos\theta_i - n_2 \cos\theta_t}{n_1 \cos\theta_i + n_2 \cos\theta_t}, \tag{10.}$$

$$t_s = \frac{2n_1 \cos\theta_i}{n_1 \cos\theta_i + n_2 \cos\theta_t}, \tag{11.}$$

$$r_p = \frac{n_2 \cos\theta_i - n_1 \cos\theta_t}{n_2 \cos\theta_i + n_1 \cos\theta_t}, \tag{12.}$$

$$t_p = \frac{2n_1 \cos\theta_i}{n_2 \cos\theta_i + n_1 \cos\theta_t} \tag{13.}$$

13. PIAA

[Notebook: How do you make a PIAA surface? And show improved performance w.r.t. standard lyot coronagraph.]

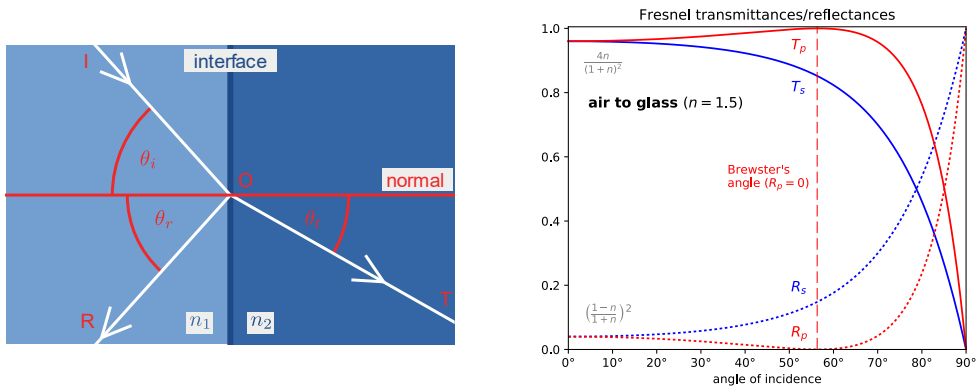

Figure 4

Phase Amplitude Induced Apodization (PIAA) remaps the telescope pupil such that a star on the optical axis forms a Gaussian PSF (Guyon 2003; Guyon et al. 2005, 2014). A focal plane mask that is apodized to match the Gaussian PSF then suppresses the on-axis starlight, while planets images off the central axis are minimally attenuated by the gaussian mask. The reimaged off axis sources have strong aberrations that are radially field dependent, significantly decreasing the Strehl ratio of these sources. A reimaging system that reverses the first PIAA optics then reforms a final focal plane image with all off axis sources forming diffraction limited images.

Results from the laboratory demonstration of a PIAACMC coronagraph with a segmented aperture, (Marx et al. 2021).

integrated WFS and coro with PIAACMC (Haffert et al. 2023a).

Laboratory demonstration of high contrast with the PIAACMC coronagraph on an obstructed and segmented aperture (Belikov et al. 2022).

Figure: SH can PIAA simulations for ELT,TMT or GMT (quick!)

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14. Coronagraphic wavefront sensing

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coronagraphic phase diversity COFFEE jaen paul sauvage 2012 (Sauvage et al. 2012).

Perspectives on phase rertrieval and phase diversity in astronomy Gonsalves (2014) and the classic Gerchberg-Saxton algorithm (Gerchberg & Saxton 1972).

A review article on phase diversity and focal plane WFS. (Fienup 2013)

Gary Ruane 2023 integrated wfs and coronagraphs (Ruane et al. 2023)

Emiel Por for PAPLC (Por 2020)

cite Por 2023 or 2021 SPIE Zernike WFS and PAPLC integrated. (Pourcelot et al. 2022, 2023)

HiCAT dark zone demonstration (Soummer et al. 2022).

SCAR corongaraph using the fiber as a filter (Haffert et al. 2020)

Direct imaging fiber nulling from Mawet et al. (2017) and Eugene Serabyn's paper on fiber nulling from Serabyn & Mennesson (2006).

Using Fresnel zones to make a circular null around the star (Angel et al. 1986) first time you could detect a Jupiter around a sun-like star.

Figure: Standard Lyot coronagraph layout

Figure: SH: segments, and segment piston errors, impact on dark hole. increase RMS with a slider, then see the dark hole filled in.

Figure: ground based coronagraphic images - real would be better, put in SH MagAOX

15. Focal plane wavefront sensing

A unified formalism for HCI algorithms was presented in Give'On (2009); Give'On et al. (2011)

???not sure what papers for this: traub 2022/2024 intensity nulling

SPHERE papers:

FIGURE: Potier 2022 see Figure 5 but SH has done it, can do a QSS at 900nm. working to shorter wavelengths

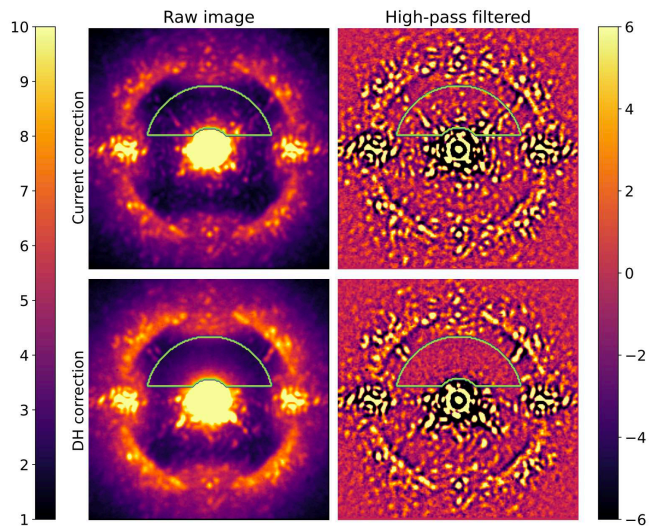

Fig. 1. Normalized intensity images in linear scale obtained before (*top row*) and after (*bottom row*) the on-sky DH correction. *Left column*: Raw image with bad pixels removed. *Right column*: Same as for the left column, but high-pass filtered using a Gaussian kernel with a standard deviation of $0.57\lambda_0/D$. The images are normalized by the maximum of the off-axis PSF, and multiplied by 10^5 and 10^6 for the left and right images, respectively. The corrected region is encircled in green.

Figure 5

[Notebook: SH: get this from HCIpy Simple dark hole generation with predefined shapes - can use the new autodiff approach.]

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16. Rejected light wavefront sensing

Visible extreme AO on ELTs for Prox b: (Fowler et al. 2023)

PAPLC type WFS....

Integrated WFS from Ruane (Ruane et al. 2020).

Talk about FAST SCC from Ben Gerard that used rejected light from coronagraph (Gerard et al. 2018).

Lyot LOWFS - Guyon, Garima Singh

MagAOX paper.... Mcleod et al. 2023

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17. PAPLC

Emiel came up with concept, 2019, has unpub paper and SPIE talk...

Haffert 2023 paper

SCOOB 2021 AZ space test bed using a knife edgs SPIE proceeding. PAPLC.

Figure: SH show unpublished on sky results with PAPLC.

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18. Segmented mirrors

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Larger telescopes are required to increase the spatial resolution of the imaging camera. The cost of a telescope goes as $D^{1.7-1.8}$ (Stahl & Allison 2020) for both, and for space only there is Stahl (2010).

Monolithic mirror telescopes are ultimately limited by their transport from manufacturing point to the observatory location, and so segmented telescope designs are used for diameters greater than 8m on the ground. Segmented mirror designs are necessary for space telescopes so that the primary mirror can be folded into the faring of rockets.

These segmented mirror designs use active controls and mechanical adjustments to align the mirrors onto the ideal primary mirror surface. This makes segmented mirror designs susceptible to temporal drifts in their alignment, leading to the generation of aberrations in the focal plane that are in the scientific region of interest.

Wavefront sensing and subsequent correction of these aberrations is therefore an important part of high

contrast imaging. Algorithms such as COFFEE have been demonstratred for the JWST segmented primary pupil geometry (Leboulleux et al. 2020).

Lucie has two papers on redundant apodization for DI of exoplanets - paper I is (Leboulleux et al. 2022a) and paper II is Leboulleux et al. (2022b).

Wavefront tolerances of space-based segmented telescopes at very high contrast: Experimental validation by Laginja et al. (2022)

A discussion of the latest limits reached by laboratory coronagraphs is in Mennesson et al. (2024).

HDFS on the GMT is described by Haffert et al. (2022).

Hedglen 2022 Exp valid of piston control with pyramid WFS is in Bertrou-Cantou et al. (2023) with the A and A paper detailing it in Confusion in differential piston measurement with the pyramid wavefront sensor (Bertrou-Cantou et al. 2022).

GMT: GMT segmented wavefront control Quiros-Pacheco SPIE phaseing for GMT (Quirós-Pacheco et al. 2022)

ELT: MELT test bed ZEUS experiment, APE experiment, both phasing experiments form ELT early/mid 2004-2006.

The MELT testbed. MELT: an optomechanical emulation testbench for ELT wavefront control and phasing strategy (Pfrommer et al. 2018)

ZEUS: a cophasing sensor based on the Zernike phase contrast method: (Dohlen et al. 2006).

On-sky Testing of the Active Phasing Experiment Gonté et al. (2009).

Keck telescope

Keck phasing: The original phasing algorithm for Keck mirror segments in Chanan et al. (1998) and the second paper in Chanan et al. (2000).

Keck mirror phasing with ZWFS in van Kooten et al. (2022) and using a Vector ZWFS in Salama et al. (2024).

On-sky verification of Fast and Furious focal-plane wavefront sensing: Moving forward toward controlling the island effect at Subaru/SCExAO with Bos et al. (2020).

petal mode problem

MAK TODO:

LOW WIND EFFECT on SPHERE (Sauvage et al. 2016) and discussions on mitigating it are described in Milli et al. (2018).

You can use fast and furious to measure it (Wilby et al. 2018) which was verified at Subaru/SCExAO.

Several mitigation strategies there with FPWFS for that (Vievard et al. 2019).

Petal mode problem for ELT: due to the spiders, they fragment the pupil into unsensed separate petals because the thickness of the secondary support is greater than r0.

Even monolithic mirrors have this problem too with thick enough secondary support structures, known as the island effect (Leboulleux et al. 2022a,b)

19. Integrated optics

Optics change the complex amplitudes of wavefronts as they propagate through coronagraphs. Macro optics, typically many thousands of times larger than the wavelength of light they shape, require precise and stable optomechanical components to accurately modify these wavefronts. Integrated optics enable direct manipulation of the complex electric fields at the scale of the wavelengths used. Miniaturisation of previously discrete macro optics and their manufacture within a single homogeneous substrate removes both the requirement for separate optomechanical alignment and temperature related effects. Early examples include beam combiners for optical astronomical interferometers (for example the IOTA/IONIC beam

combiner Berger et al. 2001) and photonic lanterns, see Leon-Saval et al. (2010) and references therein.

PIMMS: photonic integrated multimode microspectrograph from Bland-Hawthorn et al. (2010).

Advances in micron scale manufacture enable the fabrication of optical waveguides and optical cavities enabling optical solutions in the full 3D propagation regime (UGH! NEED TO TIDY UP).

Efficient injection from large telescopes into single-mode fibres: Enabling the era of ultra-precision astronomy (Jovanovic et al. 2017)

Photonic optimised coronagraphs and Integrated photonic-based coronagraphic systems for future space telescopes (Desai et al. 2023a).

Bragg gratings to get modulate diffraction (Fagginger Auer & Keller 2024).

Barnaby Norris photonic lanterns (Norris et al. 2022) and on-sky demo (Norris et al. 2020a) and nature one Norris et al. (2020b), FPWFS with photonic lanterns (Norris et al. 2020c) and their theory (Lin et al. 2022).

Nulling with a mode selective photonic lantern 2022 (Xin et al. 2022).

Phase-apodized-pupil Lyot Coronagraphs for Arbitrary Telescope Pupils (Por 2020) SCAR paper I (Por & Haffert 2020)

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20. Astrophotonics and QOD

No idea here!

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21. Quantum optimal detection

Quantum optical detection (Lu et al. 2018) enables the distinction between two incoherent point sources within the classical Rayleigh diffraction limit. This can be done by a specific linear optic reformatting of the wavefront followed by a photon counting detector. More specifically, for exoplanet detection where the separation is smaller than the diffraction limit and the flux ratio much smaller than one, for thermalised incoherent sources (e.g. a star and a planet) you test for photons not being distributed in a point symmetric way (e.g. Huang & Lupo 2021).

Desai et al. (2023b) on deriving this for Achieving Quantum Limits of Exoplanet Detection and Localization.

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22. Post-processing (its no longer a photon)

[Notebook: KLIP with removing varying number of modes and annular rings.]

2006 with ADI Marois (Schneider 2000 HST roll subtraction)

Dual band imaging/mathane imaging Beth but also french guy before..., halpha imaging Laird C

TRIDENT Marois using three bands, into standard SDI with IFUs.

High resolution SDI/molecule mapping, Snellen, really high

Polarization differential Imaging:

Coherence Differential Imaging: with MKIDS and Ben Mazin VAMPIRES, done it on SPHERE. Gladyz paper. Remove QSS. Codona, Codona and Kenworthy 2015 (hah!).

Figure: show example of post processing. pick favourite image from a paper!

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DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

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To achieve the scientific results presented in this article we made use of the *Python* programming language[3], especially the *SciPy* (Virtanen et al. 2020), *NumPy* (Oliphant 2006), *Matplotlib* (Hunter 2007), *emcee* (Foreman-Mackey et al. 2013), and *astropy* (Astropy Collaboration et al. 2013, 2018) packages.

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